



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 12 2010 ASSESSMENT TASK 3

MATHEMATICS

(EXTENSION 1)

Time Allowed - *two hours*

WEIGHTING 20% towards final result

Outcomes referred to: P2, P4, P5, P6, P7, P8, H1, H2, H4, H5, H6, H7, H9

INSTRUCTIONS:

1. Attempt **ALL** questions.
2. Show all necessary working.
3. **Begin** each question on a **new page**.
4. Each question is of equal value. Mark values are shown beside each part.
5. Non-programmable silent Board of Studies approved calculators are permitted.
6. If requested, additional writing sheets may be obtained from the examinations supervisor upon request.
7. A double sided A4 page of notes is permitted to be referred to throughout this task.

Question 1.(12marks) (Start this question on a New page)**Marks**

- (a) Find the acute angle between the lines $y = 5x - 1$ and $6x - 3y + 1 = 0$.

Give your answer correct to the nearest minute.

3

- (b) Solve $\frac{3}{x-2} > 4$.

3

- (c) Solve $x^2 \leq 9x$ and draw your answer on a number line.

3

- (d) The angle between $y = mx + 1$ and $x - 2y = 0$ is 45° .

Find the possible value(s) of m .

3**Question 2.(12marks) (Start this question on a New page)**

-
- (a) Find the coordinates of the point that divides the join of $(-3, 2)$ and $(5, 7)$ externally in the ratio 1:3.

2

- (b) The position of a particle moving along the x axis is given by:

$$x = \frac{t^3}{3} - 3t^2 + 8t \quad \text{where } x \text{ and } t \text{ are measured in centimetres and seconds}$$

respectively. Find:

- (i) expressions for the velocity and acceleration of the particle.

2

- (ii) when and where the particle first comes to rest.

2

- (iii) the distance travelled by the particle in the first 3 seconds.

2

- (c) Evaluate $\sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}(1)$

2

- (d) Find $\int \frac{2}{\sqrt{4-25x^2}} dx$

2

Question 3.(12marks) (Start this question on a New page)

(a) Solve for $0^\circ \leq \theta \leq 360^\circ$

(i) $2 \sin 2\theta = 1$ 2

(ii) $8 \sin^2 \theta - 2 \cos^2 \theta = \cos 2\theta$ (to nearest minute) 4

(b) (i) Show that $\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$ 2

(ii) Hence find the value of $\tan 22\frac{1}{2}^\circ$ in simplest exact value. 2

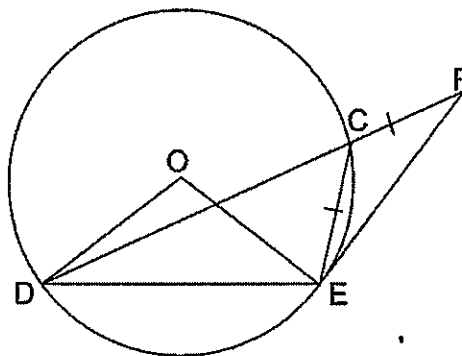
(c) Find the derivative of $\cos^{-1}\left(\frac{x}{3}\right)$ 2

Question 4.(12marks) (Start this question on a New page)

- (a) In the figure, FE is a tangent to the circle, centre O.
D and F are joined so that EC = CF.
(i) If $\angle CEF = x$, prove that $DE = EF$.
(ii) Express $\angle DOE$ in terms of x .

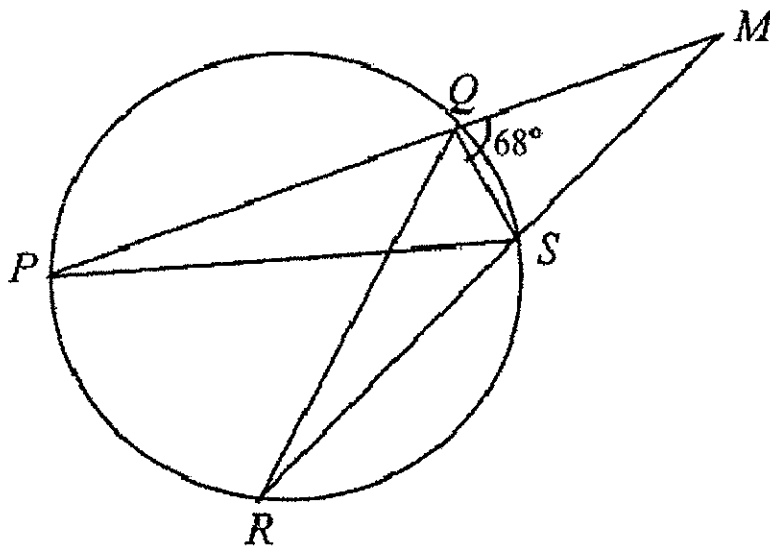
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2



- (b) In the diagram, $MQ = MS$ and $\angle MQS = 68^\circ$. Copy or trace this diagram and prove that $MP = MR$.

4



- (c) (i) Show that the point P(-2,0) lies on the line joining A(7,-3) and B(-5,1)
- (ii) Find the ratio in which it divides the line.

4

Question 5.(12marks) (Start this question on a New page)

- (a) Forty five percent of a population are of blood group O, 40% are of blood group A and the remainder are of neither group O nor group A.

Three people are chosen at random from the population

By drawing a tree diagram, or otherwise, find the probability that

- (i) all three people are of blood group A; 1
 - (ii) two of the people are of blood group A and the other is group O; 1
 - (iii) there is one person each of group O, group A, and neither group O nor group A. 2
- (b) The amount A grams of a given carbon isotope in a dead tree trunk is given by

$$A = A_0 e^{-kt}$$

where A_0 and k are positive constants, and where time t is measured in years from the death of the tree.

- (i) Show that A satisfies the equation $\frac{dA}{dt} = -kA$, 1
 - (ii) Find the value of k if the amount of isotope present is halved every 5500 years. 2
 - (iii) For a particular dead tree trunk the amount of isotope is only 15% of the original amount in the living tree. How long ago did the tree die? 2
- (c) Sketch the graph of $y = 2 \sin^{-1}\left(\frac{x}{3}\right)$ indicating clearly its domain and range. 3

Question 6.(12marks) (Start this question on a New page)

- (a) If $\tan \frac{\theta}{2} = t$, show that

$$\frac{\sin \theta + \sin \frac{\theta}{2}}{1 + \cos \theta + \cos \frac{\theta}{2}} = t$$

4

- (b) For the function $y = x^2 - 2x + 1$,

(i) find a suitable domain such that this function has an inverse.

(ii) find the equation of this inverse and state its range.

4

- (c) Prove by Mathematical induction that $3^{2n} - 1$ is divisible by 8, where n is a positive integer.

4

Question 7.(12marks) (Start this question on a New page)

- (a) Find $\frac{d}{dx}(x \tan^{-1} x)$ and hence show that

$$\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \frac{1}{2} \log_e 2 \quad 3$$

- (b) Find k if $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{5} = \tan^{-1} k$ 3

- (c) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

- (i) Derive the equation of the chord joining P and Q . 1

- (ii) If PQ passes through $(2a, 0)$ show that $p + q = pq$. 1

- (iii) Hence show that the locus of M , the mid point of PQ is a parabola. 2

- (iv) Find the vertex and focus of the locus of M . 2

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Year 12 Ex 1.

Question 1

a) $m_1 = 5$

$m_2 = 2$

① (one correct)

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{5 - 2}{1 + 5 \times 2} \right| = \frac{3}{11}$$

① (correct fraction)
① (from above)

$$\theta = 15^\circ 15'$$

① (correct angle)
① (from above)

b) $\frac{3}{x-2} > 4$

$x \neq 2$

① (must have)

using zero $-\frac{3}{2} > 4$

$$3 = 4x - 8$$

$$11 = 4x$$

$$x = \frac{11}{4}$$

① (or equivalent)

check.

$$2 < x < \frac{11}{4}$$

① (soln)

c) $x^2 \leq 9x$

$$x^2 - 9x \leq 0$$

$$x(x-9) = 0$$

$$x = 0 \text{ or } 9$$

assume $\neq 0$

① (or equivalent)

① (soln).

$$0 \leq x \leq 9$$

① soln.

check $1^2 \leq 9$ works.

d) $m_1 = m$

$m_2 = \frac{1}{2}$

$\theta = 45$

$$\tan 45 = \left| \frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} \right|$$

① (subst)

$$1 = \frac{m - \frac{1}{2}}{1 + \frac{1}{2}m}$$

or

$$1 = \frac{-(m - \frac{1}{2})}{1 + \frac{1}{2}m}$$

$$1 + \frac{1}{2}m = m - \frac{1}{2}$$

$$1 + \frac{1}{2}m = -m + \frac{1}{2}$$

$$1\frac{1}{2} = \frac{1}{2}m$$

$$\frac{3}{2}m = -\frac{1}{2}$$

$$m = 3$$

① soln

$$m = -\frac{1}{3}$$

① soln.

Question 2.

a) $(-3, 2) \quad (5, 7) \quad -1:3$

$$x_3 = \frac{3 \times -3 + -1 \times 5}{-1 + 3}$$

$$= \frac{-14}{2}$$

$$= -7$$

$$y_3 = \frac{3 \times 2 + -1 \times 7}{-1 + 3}$$

$$= \frac{-1}{2}$$

① (correct sub in x & y)

① (Soln)

$$(-7, -\frac{1}{2})$$

b) $x = \frac{t^3}{3} - 3t^2 + 8t$

i) $\frac{dx}{dt} = v = t^2 - 6t + 8$

①

$$\frac{d^2x}{dt^2} = a = 2t - 6$$

①

ii) rest $v = 0$

$$t^2 - 6t + 8 = 0$$

$$(t - 4)(t - 2) = 0$$

when $t_1 = 4$ or 2

①

where $x_4 = \frac{64}{3} - 48 + 32$

$$x_2 = \frac{8}{3} - 12 + 16$$

$$= 21\frac{1}{3} - 48 + 32$$

$$= 2\frac{2}{3} - 12 + 16$$

$$= 5\frac{1}{3}$$

$$= 6\frac{2}{3}$$

①

(if only one answer for t & correct x ①)

iii) $t = 0 \quad x = 0$

$$t = 3 \quad x = 9 - 27 + 24 = 6$$

① (for 6.)

$$t = 2 \quad x = 6\frac{2}{3}$$

$$\text{Distance} = \int_{t=0}^{t=2} 6\frac{2}{3} + \int_{t=2}^{t=3} \frac{2}{3} = 7\frac{1}{3}$$

② (for $7\frac{1}{3}$)

c) $\sin^{-1}(\frac{1}{2}) + \cos^{-1}(1) = \frac{\pi}{6} + 0 = \frac{\pi}{6}$

① (for either correct)

① (Answer)

d) $\int \frac{2}{\sqrt{4-25x^2}} dx = 2 \int \frac{dx}{\sqrt{4-25x^2}}$

$$= 2 \sin^{-1}(\frac{5x}{2}) \times \frac{1}{5}$$

① (without $\frac{1}{5}$)

$$= \frac{2}{5} \sin^{-1}(\frac{5x}{2})$$

① (Ans)

Question 3

a) Solve $0 \leq \theta \leq 360$

$$2 \sin 2\theta = 1$$

$$\sin 2\theta = \frac{1}{2}$$

$$2\theta = \sin^{-1}\left(\frac{1}{2}\right) = 30, 150, 390, 510 \quad (2 \text{ Answers})$$

$$\theta = 15, 75, 195, 255 \quad (1 \text{ Ans})$$

b) ii) $8 \sin^2 \theta - 2 \cos^2 \theta = \cos 2\theta$

$$= \cos^2 \theta - \sin^2 \theta \quad (1)$$

$$9 \sin^2 \theta - 3 \cos^2 \theta = 0$$

(or equiv.)

$$9 \sin^2 \theta = 3 \cos^2 \theta$$

$$\div \sin^2 \theta \quad (1)$$

$$3 \tan^2 \theta = 1$$

$$\tan^2 \theta = \frac{1}{3}$$

$$\tan \theta = \pm \frac{1}{\sqrt{3}}$$

$$(1) \left(\begin{smallmatrix} + \\ - \end{smallmatrix} \right)$$

$$\theta = 30, 150, 210, 330$$

$$(1) \text{ (four soln)}$$

(only 2 soln 3 marks)

b) i) $\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$

$$\frac{1 - (\cos^2 x - \sin^2 x)}{1 + (\cos^2 x - \sin^2 x)}$$

$$(1)$$

$$\frac{\sin^2 x + \sin^2 x}{\cos^2 x + \cos^2 x}$$

$$1 - \sin^2 x = \cos^2 x$$

$$1 - \cos^2 x = \sin^2 x$$

$$\frac{2 \sin^2 x}{2 \cos^2 x}$$

$$= \tan^2 x$$

$$(1)$$

$$\tan x = \frac{\sin x}{\cos x}$$

(soln)

ii) $\tan^2(22\frac{1}{2}^\circ) = \frac{1 - \cos 45}{1 + \cos 45}$

$$= \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$$

$$(1) \text{ (Simplify)}$$

$$\tan(22\frac{1}{2}) = \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2} + 1}}$$

$$(1) \text{ soln}$$

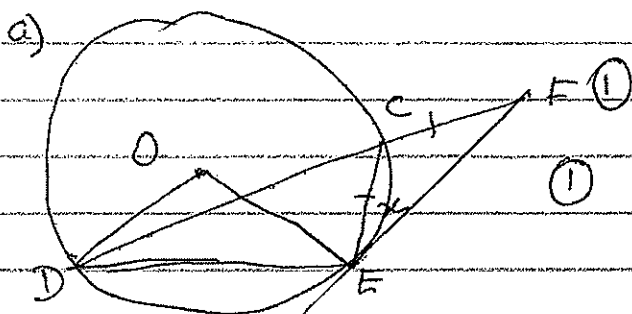
(ignore if went further)

c) $\frac{d}{dx} \left(\cos^{-1}\left(\frac{x}{3}\right) \right) = \frac{-1}{\sqrt{9 - x^2}}$

$$(2) \text{ soln}$$

(1) (no negative sign)

Question 4



i) $\angle CEF = x$ Prove $DE = EF$

$\angle CFE = x$ (Isosceles Δ)

$\angle CDE = \angle CEF$ (angles in alternate segment from tangent)

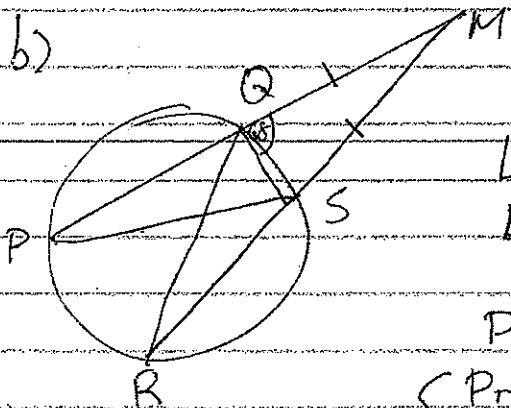
$\therefore DE = EF$ (Base angles are equal)

ii) $\angle DEF = 90^\circ$ (angle from centre to tangent = 90°)

$\angle OEC = 90 - x$

$\angle DCE = 2x$ (external angle of Δ)

$\therefore \angle DOE = 4x$ (angle at centre is twice angle at circumference)
(2 for equivalent solution)



Prove $MP = MR$

$\angle MQS = 68 = \angle MSQ$ (Isosceles Δ)

Let $\angle PQR = x = \angle RSP$ (angles on same arc are equal)

$\angle PQS = 180 - 68 = 112^\circ = \angle RSQ$ (str lines)

Prove $\Delta PQS \equiv \Delta RSQ$

Proof 1. $\angle PQS = \angle RSQ = 112^\circ$ (as above)

2. QS is common

3. $\angle P = \angle R$ (angles standing on same arc are equal)

$\therefore \Delta PQS \equiv \Delta RSQ$ (AAS)

$PQ = RS$ (matching sides)

① $\therefore PM = QM + PQ = MS + SR = MR$

c) i) $m_{AB} = \frac{1-3}{-5-7} = \frac{4}{-12} = -\frac{1}{3}$

line AB $y - 3 = -\frac{1}{3}(x - 7)$

$3y + 9 = -x + 7$

$x + 3y + 2 = 0$

check $P(-2, 0)$ lies on $-2 + 0 + 2 = 0 \checkmark$

$\therefore P$ lies on AB

① (line)

① (subst)

Q4c ii)

$$x_3 = \frac{mx_2 + nx_1}{m+n}$$

$$y_3 = \frac{my_2 + ny_1}{m+n}$$

$$-2 = \frac{-5m+7n}{m+n}$$

$$0 = \frac{m-3n}{m+n}$$

$$-2m-2n = -5m+7n$$

$$0 = m-3n$$

$$3m = 9n$$

$$m = 3n$$

$$m = 3n$$

① (eqn)

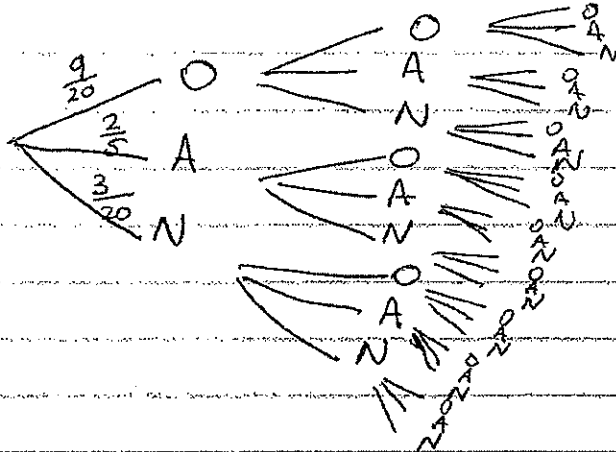
$\therefore m$ is 3 times bigger

Ratio 3:1

① (correct ratio)

Question 5

a) 45% is O 40% is A 15% neither



$$i) P(3A) = \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} = \frac{8}{125} \quad (1)$$

$$ii) P(2A, 0) = \frac{2}{5} \times \frac{2}{5} \times \frac{9}{20} \times 3 = \frac{108}{500} = \frac{27}{125} \quad (1)$$

$$iii) P(1 \text{ of each}) = \frac{2}{5} \times \frac{9}{20} \times \frac{3}{20} \times 6 = \frac{81}{500}$$

(3 numbers mult) (1) for 6 ways.

$$b) A = A_0 e^{-kt} \quad (1)$$

$$i) \frac{dA}{dt} = -k A_0 e^{-kt} \quad \text{from (1)} \quad (1)$$

$$= -k A$$

$$ii) t = 5500 \quad A = \frac{1}{2} A_0$$

$$\frac{1}{2} A_0 = A_0 e^{-k \times 5500}$$

$$\frac{1}{2} = e^{-5500k} \quad \text{log both sides} \quad (1)$$

$$\ln \frac{1}{2} = \ln e^{-5500k} = -5500k$$

$$k = \frac{\ln \frac{1}{2}}{-5500} = 1.260267 \times 10^{-4} \quad (1)$$

$$\approx 1.26 \times 10^{-4} \text{ or } 0.000126$$

$$iii) \frac{3}{20} A_0 = A_0 e^{-kt}$$

$$\frac{3}{20} = e^{-kt}$$

$$\ln\left(\frac{3}{20}\right) = \ln e^{-kt} = -kt \quad (1)$$

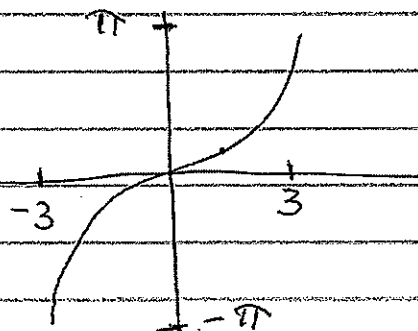
$$t = \ln\left(\frac{3}{20}\right) \div (-k) = 15053 \text{ yrs} \quad (1)$$

$$\approx 15000 \text{ or } 15100$$

$$c) y = 2 \sin^{-1}\left(\frac{x}{3}\right)$$

(1) showing x values

(1) showing y values.



Question 6

a) $\tan \frac{\theta}{2} = t$.

$$\sin \theta = \sin \left(\frac{\theta}{2} + \frac{\theta}{2} \right) = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \quad (1)$$

$$\cos \theta = \cos \left(\frac{\theta}{2} + \frac{\theta}{2} \right) = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \quad (1)$$

$$\frac{\sin \theta + \sin \frac{\theta}{2}}{1 + \cos \theta + \cos \frac{\theta}{2}} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{1 + \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} + \cos \frac{\theta}{2}}$$

$$= \frac{\sin \frac{\theta}{2} (2 \cos \frac{\theta}{2} + 1)}{\cos^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} + \cos \frac{\theta}{2}}$$

$$1 - \sin^2 \frac{\theta}{2} = \cos^2 \frac{\theta}{2}$$

$$= \frac{\sin \frac{\theta}{2} (2 \cos \frac{\theta}{2} + 1)}{\cos \frac{\theta}{2} (2 \cos \frac{\theta}{2} + 1)} \quad (1)$$

$$= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} = \tan \frac{\theta}{2} = t. \quad (1)$$

(or equivalent).

b) $y = x^2 - 2x + 1 = (x-1)^2$

i) Vertex (1, 0)

(1) Vertex

Domain $x \geq 1$ (1) or $x \leq 1$

ii) Inverse $x = y^2 - 2y + 1$

$$x = (y-1)^2$$

$$\pm \sqrt{x} = y - 1$$

$$y = 1 \pm \sqrt{x}$$

(1) (must show both)

if $x \geq 1$ then $y = 1 + \sqrt{x}$ (1)

c) Prove $8 \mid 3^{2n} - 1$

Prove true for $n=1$

$$8 \mid 3^2 - 1 = 9 - 1 = 8$$

$$\therefore 8/8 = 1 \text{ true for } n=1$$

Assume true for $n=k$.

$$8 \mid 3^{2k} - 1$$

$$8M = 3^{2k} - 1$$

$$3^{2k} = 8M + 1$$

Assume answer true for some positive integer M

Q6 c) Prove true for $n = k+1$

$$8 \mid 3^{2(k+1)} - 1$$

$$3^{2k+2} - 1$$

$$= 3^2 \cdot 3^{2k} - 1$$

$$\text{now } 3^{2k} = 8M+1$$

$$\therefore 8 \mid 9(8M+1) - 1$$

$$72M+9-1$$

$$72M+8$$

$$8 \mid 8(9M+1)$$

$\therefore$ true for $n = k+1$

$\therefore$ true for $n=1, 2, \dots$ true for all positive integer of $n \geq 1$.

Question 7

$$a) \frac{d}{dx}(x \tan^{-1} x) = x \cdot \frac{1}{1+x^2} + 1 \cdot \tan^{-1} x$$

$$= \frac{x}{1+x^2} + \tan^{-1} x \quad (1)$$

Now integrate both sides.

$$\text{Now } \int \frac{d}{dx}(x \tan^{-1} x) dx = \int \frac{x}{1+x^2} dx + \int \tan^{-1} x dx$$

$$[x \tan^{-1} x]_0^1 = \frac{1}{2} [\ln(1+x^2)]_0^1 + \int_0^1 \tan^{-1} x dx \quad (1)$$

$$\tan^{-1} 1 - 0 = \frac{1}{2} (\ln 2 - \ln 1)$$

$$\tan^{-1}(1) = \frac{1}{2} \ln 2 + \int_0^1 \tan^{-1} x dx$$

$$\int_0^1 \tan^{-1} x dx = \tan^{-1}(1) - \frac{1}{2} \ln 2 = \frac{\pi}{4} - \frac{1}{2} \ln 2 \quad (1)$$

$$b) \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) = \tan^{-1}(k)$$

$$\tan\left(\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right)\right) = k \quad (1)$$

$$\tan(\alpha + \beta) = k$$

$$k = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{\tan\left(\tan^{-1}\left(\frac{1}{3}\right)\right) + \tan\left(\tan^{-1}\left(\frac{1}{5}\right)\right)}{1 - \tan\left(\tan^{-1}\left(\frac{1}{3}\right)\right) \tan\left(\tan^{-1}\left(\frac{1}{5}\right)\right)}$$

$$= \frac{\frac{1}{3} + \frac{1}{5}}{1 - \frac{1}{3} \times \frac{1}{5}} = \frac{\frac{8}{15}}{1 - \frac{1}{15}} = \frac{8}{14} = \frac{4}{7} \quad (1)$$

$$c) P(2ap, ap^2) \quad Q(2aq, aq^2)$$

$$i) m_{PQ} = \frac{aq^2 - ap^2}{2aq - 2ap} = \frac{a(q-p)(q+p)}{2a(q-p)} = \frac{q+p}{2}$$

$$y - ap^2 = \left(\frac{q+p}{2}\right)(x - 2ap) \quad (1)$$

$$2y - 2ap^2 = (q+p)x - 2apq - 2ap^2$$

$$(q+p)x - 2y - 2apq = 0$$

$$ii) (2a, 0)$$

$$(q+p) \times 2a - 0 - 2apq = 0$$

$$2aq + 2ap = 2apq \quad \div 2a$$

$$q+p = pq \quad (1)$$

Q7 c iii) Mid pt. $\left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right)$

$$x = a(p+q)$$

$$\frac{2x}{a} = p+q$$

$$y = \frac{ap^2+aq^2}{2}$$

$$= \frac{a}{2}(p^2+q^2)$$

$$y = \frac{a}{2}[(p+q)^2 - 2pq] \quad (1) \quad \text{(getting eqn.)}$$

$$\text{since } pq = p+q$$

$$y = \frac{a}{2}[(p+q)^2 - 2(p+q)]$$

$$y = \frac{a}{2}\left[\left(\frac{x}{a}\right)^2 - 2\left(\frac{x}{a}\right)\right] \quad (1) \text{ soln.}$$

iv)

$$y = \frac{a}{2}\left[\frac{x^2}{a^2} - \frac{2x}{a}\right]$$

$$y = \frac{x^2}{2a} - x$$

$$2ay = x^2 - 2ax$$

$$2ay = (x-a)^2 - a^2$$

$$2ay + a^2 = (x-a)^2$$

$$2a\left(y + \frac{a}{2}\right) = (x-a)^2$$

Vertex $\left(a, -\frac{a}{2}\right)$ (1) for Equation

$$x^2 = 4ay$$

but only $2a$ in eqn.

focal length $\frac{a}{2}$ not a

Focus $(a, 0)$. (1)